

Challenge from Microsoft

Hand Gesture Classification on Edge Devices

This project focuses on building a compact, reliable hand-gesture classifier for edge devices using TA-provided hand detection and landmark extraction. The classifier must take only a cropped hand image and 21 landmark coordinates as input, recognize five designated HaGRIDv2 command gestures, and output N/A for all other gestures to prevent false triggers under real-world conditions. The goal is to design a system for the real world with noisy input.

1. Fixed Pipeline & Input Constraints

To ensure a fair baseline, the preprocessing module (Hand Detection & Landmarking) will be provided by the TAs. Students are not required to survey hand detection models. Your task is to design and train the downstream **Gesture Classifier**.

The preprocessing module uses **MediaPipe**. Students should refer to the provided preprocessing code for the exact input/output format. The classifier will only receive the output of this preprocessing module.

Strict Input Constraints: Your model and inference script may **ONLY** accept the following two inputs. Full-frame raw images are strictly prohibited.

1. Gesture Cropped Image: The localized RGB image of the hand.
2. Joint Landmarks: 21 structural hand landmark coordinates.

2. Target Classes (HaGRIDv2)

The official dataset for this project is **HaGRIDv2**. We designate **5 specific gestures** (fist, like, ok, one, palm) as valid control commands.

- **Valid Commands (Class 1~5):** Must be classified accurately.
- **N/A State (Class 0):** Any gesture not belonging to the 5 designated classes, as well as any unknown actions, must be output as N/A.

For the HaGRIDv2-based N/A samples, we use one-handed non-target gestures. Two-handed gesture categories are excluded. We use the lightweight **HaGRIDv2 512px** version for testing.

The target gestures follow the official HaGRIDv2 sample images. Target gestures are evaluated ONLY under standard poses. Ambiguous, distorted, or non-standard poses must be treated as N/A.

3. Submission Format

You must submit executable **Python code** (`inference.py`) and **your model weights** through the provided Google Form. Each team must submit a single zip file with the following structure:

```
team_X.zip
├── inference.py
├── model/
│   └── model weights or checkpoint files
├── requirements.txt      # inference-only dependencies
└── README.md            # environment and brief usage notes
```

The submitted `inference.py` must be placed at the top level of the zip file and must implement the following interface:

```
# You must implement and submit this interface

def predict(cropped_img: np.ndarray, landmarks: np.ndarray) -> int:
    return final_decision_class # {0,1,2,3,4,5}

# where: 0 = N/A, 1 = fist, 2 = like, 3 = ok, 4 = one, 5 = palm
# All paths in inference.py should be relative to inference.py.
```

The official evaluation will be conducted in a fresh **Google Colab** runtime. Students must ensure that their submitted code can be executed in this environment without manual modification.

 **Engineering Heuristics Allowed:** Determining the N/A class does not need to rely purely on your neural network's forward pass. You are highly encouraged to implement engineering hacks inside your Python code—such as observing probability distributions (softmax outputs), applying confidence thresholding, or writing specific edge-case logic. We do not restrict how you achieve the N/A output, as long as the final returned answer is correct.

Submission Schedule: TAs will evaluate submissions on **Monday, Wednesday, Friday, and Sunday** and update the leaderboard after each evaluation, starting from May 29th.

4. Evaluation Metrics (120 pts)

Pts	Criteria	Description
30 pts	#1 Model Size Constraint	Strict limit: ≤ 10 MB. Formula: $\text{Score} = (10 - \text{Model_Size}) * 3$. (Exceeding 10MB results in 0 points for this section).
20 pts	#2 Basic Performance (HaGRIDv2)	Test set directly sourced from HaGRIDv2 512px (5 target classes + others). <i>Note: TAs will apply reasonable augmentations (e.g., bbox jitter, blur, etc.).</i> Scoring: • Correctly predicting a 5-class target: +1 pt • False Trigger (misclassification or triggering on N/A): -2 pts Basic_Perf_Score = (RawScore / MaxRawScore) $\times$ 20
40 pts	#3 Real World Robustness	Evaluated on a custom TA-shot dataset: 50 N/A interference images (daily actions) + 50 5-class images. Scoring: • Correctly predicting a 5-class target: +1 pt • False Trigger: -2 pts Robustness_Score = (RawScore / MaxRawScore) $\times$ 40
30 pts	Presentation	Live Demo and explanation of your defense mechanisms and engineering strategies.

5. Reminder

Pretrained models trained on HaGRID / HaGRIDv2 or similar gesture recognition datasets are **NOT** allowed.

Allowed:

- ImageNet-pretrained backbones
- MediaPipe Hand Landmarker